

ATIVIDADE DE APOIO – ENSINO MÉDIO

Componente Curricular: Física

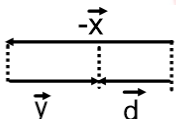
Professor (a): Rosana

Assunto (s): Diferença e Decomposição de Vetores - Resolução

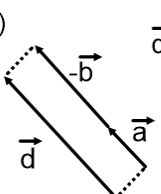
Data: Abril/25

Série: 1

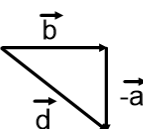
1-

a)  $\vec{d} = \vec{y} + (-\vec{x})$

$$|\vec{d}| = |-\vec{x}| - |\vec{y}| \Rightarrow |\vec{d}| = 5 - 3 \Rightarrow |\vec{d}| = 2 \text{ u.m.}$$

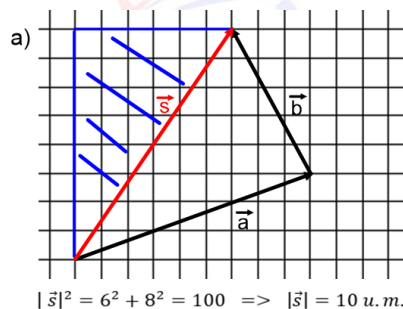
b)  $\vec{d} = \vec{a} + (-\vec{b})$

$$|\vec{d}| = |\vec{a}| + |-\vec{b}| \Rightarrow |\vec{d}| = 4 + 2 \Rightarrow |\vec{d}| = 6 \text{ u.m.}$$

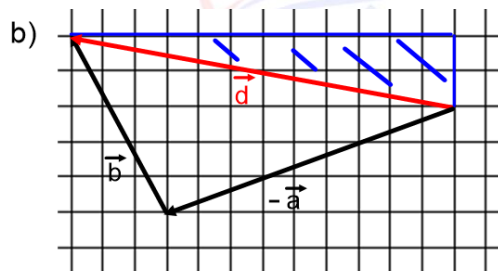
c)  $\vec{d} = \vec{b} + (-\vec{a})$

$$|\vec{d}|^2 = |\vec{b}|^2 + |-\vec{a}|^2 \Rightarrow |\vec{d}|^2 = 3^2 + 4^2 = 25 \Rightarrow |\vec{d}| = 5 \text{ u.m.}$$

2-



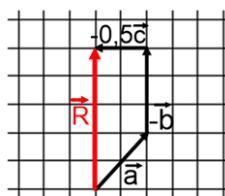
$$|\vec{s}|^2 = 6^2 + 8^2 = 100 \Rightarrow |\vec{s}| = 10 \text{ u.m.}$$



$$|\vec{d}|^2 = 12^2 + 2^2 = 148 \Rightarrow |\vec{d}| = 2\sqrt{37} \text{ u.m.}$$

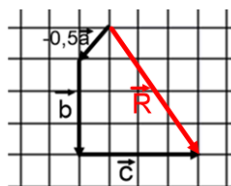
3-

a) $\vec{R} = \vec{a} - \vec{b} - 0,5 \cdot \vec{c};$



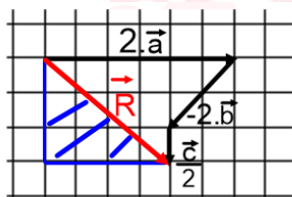
 $|\vec{R}| = 5 \text{ cm}$

b) $\vec{R} = -0,5 \cdot \vec{a} + \vec{b} + \vec{c};$



 $|\vec{R}|^2 = 3^2 + 4^2 = 25 \Rightarrow |\vec{R}| = 5 \text{ cm}$

4-



$$|\vec{R}|^2 = 3^2 + 4^2 = 25 \Rightarrow |\vec{R}| = 5 \text{ u.m.}$$

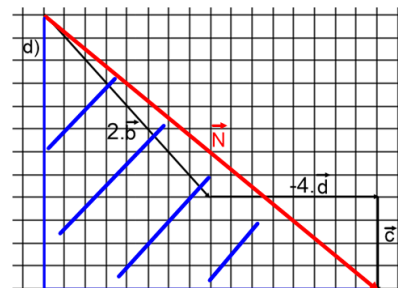
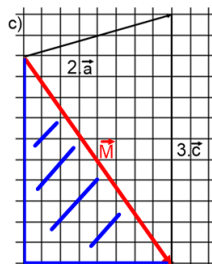
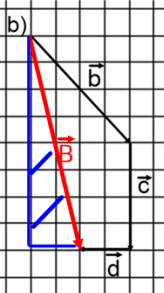
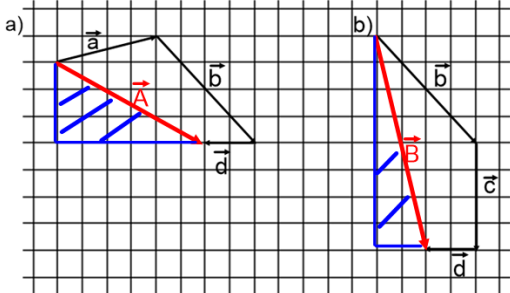
5-

a) $\vec{A} = \vec{a} + \vec{b} + \vec{d}$

b) $\vec{B} = \vec{b} + \vec{c} + \vec{d}$

c) $\vec{M} = 2\vec{a} + 3\vec{c}$

d) $\vec{N} = 2\vec{b} + \vec{c} - 4\vec{d}$



$$|\vec{M}|^2 = 10^2 + 8^2 = 164 \Rightarrow |\vec{M}| = 2\sqrt{41} \text{ u.m.}$$

$$|\vec{N}|^2 = 12^2 + 16^2 = 400 \Rightarrow |\vec{N}| = 20 \text{ u.m.}$$

a) $|\vec{A}|^2 = 3^2 + 6^2 = 45 \Rightarrow |\vec{A}| = 3\sqrt{5} \text{ u.m.}$

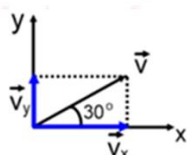
b) $|\vec{B}|^2 = 2^2 + 8^2 = 88 \Rightarrow |\vec{B}| = 2\sqrt{17} \text{ u.m.}$

6-

o vetor $\vec{v}$ tem módulo igual a 10 u.m.



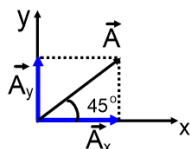
Dados: $\sin 30^\circ = \frac{1}{2}$; $\cos 30^\circ = \frac{\sqrt{3}}{2}$



* $v_x = v \cdot \cos 30^\circ \Rightarrow v_x = 10 \cdot \frac{\sqrt{3}}{2} \Rightarrow v_x = 5\sqrt{3} \text{ u.m.}$

* $v_y = v \cdot \sin 30^\circ \Rightarrow v_y = 10 \cdot 0,5 \Rightarrow v_y = 5 \text{ u.m.}$

7-



* $A_y = A_x = v \cdot \cos 45^\circ \Rightarrow A_x = 20 \cdot \frac{\sqrt{2}}{2} \Rightarrow A_x = 10\sqrt{2} \text{ cm}$